

The Theory of Optimum Currency Areas: A Survey

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IN RECENT YEARS considerable scholarly attention has focused on the question of optimum currency areas.¹ Theoretically defined, optimum currency areas may not necessarily correspond to national frontiers. Since economic nonhomogeneity exists within a single nation and since there are both large and small nations in the world, monetary independence and a flexible exchange rate for every economic region within a country, or even for every country, may not lead to desirable consequences. A separate currency for each individual would make no sense at all, because such an arrangement would reduce the world to barter. In view of this doubt about the appropriateness of flexible exchange rates for all of the existing national currencies, the question arises: What is the appropriate domain of a currency area?

There are two approaches to answering this question. The traditional approach tries to single out a crucial economic characteristic which supposedly indicates where the lines should be drawn. Contributions by Mundell (1961), McKinnon (1963a), and Kenen (1969) employ this approach. Others, however, start with the national economies,

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¹ Although Mundell (1961) was the first to write formally on the theory of optimum currency areas, the balance of payments implications of economic integration had previously received the attention of several authors, especially Balogh (1950), Scitovsky (1958), Yeager (1958), and Meade (1957). Mundell's contribution was subsequently followed by the contributions of McKinnon (1963a, 1963b, and 1973), Kenen (1969), Snider (1967), Haberler (1970), Grubel (1970), Fleming (1971), Corden (1972 and 1973), Giersch (1973), Whitman (1967), Claassen (1972), Hodgman (1972), Johnson (1971 and 1973a), Kaldor (1971), Kindleberger (1971), Griffiths (1971), Lamfalussy (1974), Oppenheimer (1973), Pearce (1974), Tower and Willett (1970 and 1975), and Williamson (1973).

taking each nation and national currency as an indivisible unit to constitute a currency area. This is a deviation from the early discussion on optimum currency areas. But a nation with numerous economically nonhomogeneous subregions can be viewed as a homogeneous entity in terms of policy attitude and national preference, and in that sense this approach also falls under the traditional approach.

The alternative approach to answering the question appears to be more helpful and capable of providing a unifying framework. It recognizes the shortcomings of theories based on a single facet of the economy and tries to evaluate costs and benefits of participating in a currency union from the point of view of the self-interest of a particular region or country. According to this approach, the optimum currency area is defined at the margin by the balance between the costs and benefits of having a common currency. Although this approach is more complex, this author finds it more useful and superior to the traditional approach.

The origins of the concept of optimum currency areas can be found in the long-standing controversial discussion of the optimal exchange rate regime. The controversy revolves around the pros and cons of fixed or flexible exchange rates in the abstract. Proponents of exchange rate flexibility, represented by Friedman (1953) and Sohmen (1969a), paid scant attention to the asymmetry among countries in the real world and left an impression that an identical case for flexibility can be made for any country. Although countries vary in size, openness to foreign trade, mobility of factors of production, and policy attitude, these proponents of exchange rate flexibility tended to treat all the national currencies as coequals. Not only proponents but also opponents of flexible exchange rates took it for granted that nations are the areas to which flexible exchange rates should be applied.

The inadequacy of this approach is now widely recognized, largely owing to recent discussions which consider biases and asymmetries among countries in the real world. On the assumption that a system of flexible exchange rates among major currencies functions well, the question of determining an optimum currency area has direct relevance to international monetary arrangements. Indeed, recent interest in optimum currency areas has been revived by the proposal for a common European currency as put forward in the Werner Report of the Commission of the European Communities (1970), and by the increased flexibility among the principal currencies of the world. Here the theory of optimum currency areas offers important considerations as to the conditions under which a country can join a currency group with posi-

tive net benefits and suggests how nations of the world should be grouped into a small number of currency zones.

In evaluating the usefulness of the optimum currency area approach, however, two considerations should be borne in mind. In the first place, it recommends a fixed exchange rate if the pros outweigh the cons from the viewpoint of national self-interest and welfare, rather than from that of global welfare, which might run counter to the former. This is why the question of an optimum world currency arrangement cannot be left to the optimum currency area approach alone. Second, the optimum currency area approach compares irrevocably fixed exchange rates with purely flexible exchange rates, while, in practice, these polar cases may not be the relevant comparison.

The discussion here is organized as follows. Section I presents a survey of original theories of optimum currency areas, with comments on each "criterion." It becomes clear that none of the single criteria of the optimum currency area so far proposed is free from difficulties. Because of these difficulties, other considerations must be introduced in order to pass judgment on a particular country regarding its exchange rate status; a variety of pros and cons on the fixing of the exchange rate need to be weighed in a manner specific to the country in question. General evaluation of theories based on a single criterion, however, is relegated to Section II, where the discussion focuses on the implications, or the possible gains and losses of having a common currency—that is, implications beyond the standard policy objectives of internal and external equilibrium. It will be seen that the traditional approach tended to concentrate on these standard policy objectives and hence neglected broader considerations that have to be taken into account in examining the desirability of participating in or forming a currency union. Finally, some further questions related to the problems of European monetary integration and the world monetary arrangement are referred to in Section III.

I. The Traditional Approach

There are both benefits and costs for a region having a currency in common with other regions, as compared to having a separate currency. In the traditional approach, the optimality of a currency area is judged with reference to a single criterion of success: specifically, achievement of the standard objectives of economic policy—full employment, price stability, and balance of payments equilibrium. The proposed single criteria are discussed below.

HIGH MOBILITY OF FACTORS OF PRODUCTION

Mundell (1961) chooses a high degree of factor mobility within the area as a basis for a unified currency area. On the assumption that a demand shift is the cause of balance of payments disequilibrium, Mundell focuses on the condition under which payments adjustment can be made with minimum burden to the regions affected by the shift. His main argument is quite simple.

Suppose that the demand shifts from the output of region *B* to the output of region *A*, where regions are defined by a high degree of mobility of factors of production, and that the standard mechanism restores the external balance through the exchange rate change. Then, if regions *B* and *A* coincide with countries *B* and *A*, a depreciation of country *B*'s currency and an appreciation of country *A*'s currency will relieve unemployment in country *B* and restrain inflation in country *A*. Therefore, a favorable case is made for flexible exchange rates. If, however, a demand shift takes place which does not correspond to national boundaries but is spread over two countries, then the two countries would encounter the following dilemma. If the central banks in both countries expand money supply in order to solve unemployment in the depressed region, they cannot solve inflation in the prosperous region; and if the central banks contract money supply in order to solve inflation in the prosperous region, they cannot cure unemployment in the depressed region. Thus, a flexible exchange rate system may solve the external imbalance between the two countries, but it fails to achieve a simultaneous solution to the problems of unemployment and inflation.

Mundell points out that this argument does not necessarily destroy the case for the flexible exchange rate system per se, but it would severely impair its relevance to national currencies; the logic of the case for flexible exchange rates can only be supported if national currencies are abandoned and regional currencies are adopted instead. If one could reorganize the world into coherent regions in Mundell's sense and create regional currencies, the second example would reduce to the first, with balance of payments adjustment between the two regions taking place through exchange rate changes. Since countries contain a variety of regions, the interregional payments adjustment problem does not seem to be adequately solved by the adjustment of national currencies. Mundell derives from this argument the principle that there must be a high degree of factor mobility in order for fixed exchange rates to be used in consistency with price stability and full employment.

Essential as it is in characterizing the optimum currency area, the

stabilization argument is not the sole consideration in determining the optimality of an area. Mundell (1961, p. 662) states:

If . . . the goals of internal stability are to be rigidly pursued, it follows that the greater is the number of separate currency areas in the world, the more successfully will these goals be attained. . . . But this seems to imply that regions ought to be defined so narrowly as to count every minor pocket of unemployment arising from labor immobility as a separate region, each of which should apparently have a separate currency!

The function of money as a medium of exchange diminishes, and the costs of currency conversion increase, as the number of currencies increases under flexible exchange rates. This point is the major argument raised by Mundell himself against the concept of regional currencies based on the stabilization argument. It is an offset to the benefit of having a small currency area, in terms of the employment problem. This conflict between the medium of exchange function and the stabilization instrument of a currency lies at the heart of almost any international monetary problem (see Section II).

Mundell puts forward another factor calling for larger size in currency areas; a small currency area necessarily implies a thin foreign exchange market, which would allow speculators to affect the market. Destabilizing speculation disrupts domestic monetary policy. Unfortunately, however, Mundell does not show how these benefits of a larger size are to be set against costs in the determination of an optimal currency area; he suggests only a simple proposition that where factor mobility exists, there is less necessity of exchange rate variations for correcting payments imbalances (arising from changing demand conditions).

Mundell's argument based on factor mobility is criticized for two reasons. First, stressing that an important distinction must be drawn between labor mobility and capital mobility, Fleming (1971) points out that internationally mobile capital as a precondition for countries to form a currency area depends on such things as the nature of the disequilibrium, the sensitivity of investment to economic activity, and the allowed time period of adjustment. For example, suppose that the demand shifts from country *B*'s products to those of country *A*, as in Mundell's first example. Suppose also that the exchange rate is rigidly maintained between the two national currencies and that payments equilibrium is restored partly by deflation (and unemployment) in country *B* and partly by inflation in country *A*. Then, the incentive to invest would decline in *B* and rise in *A*. If the investment incentive declines more than savings decline in *B*, and rises more than savings

rise in *A*, there will be an aggravation of unemployment in *B* and of inflation in *A*. In this case capital mobility worsens internal disequilibrium, although the possibility of equilibrating capital flows is also present.

Second, and more seriously, the assumption of labor mobility in Mundell's analysis is criticized by Lanyi (1969), Scitovsky (1967), Dunn (1971), and Corden (1973) on the grounds that in practice it is unlikely that sufficient interregional labor mobility can be counted on as a mechanism for payments adjustment and that even if there is mobility, the costs of movement cannot be ignored.² It is simply doubtful that the movement of working masses can be relied on as a substitute for payments adjustment when it can be assumed that they are reluctant to move even within the same country. Corden (1973, pp. 167–68) states:

Can it really be imagined that a U.K. depressed-area problem could be solved by the large-scale migration of British workers to Germany? It is conceivable; but when Britons are reluctant even to move from Scotland or Tyneside to the south, though the language is almost the same, it takes some imagination to conceive of labor mobility solving the central problem of monetary integration.

If labor mobility is to be used as a payments adjustment mechanism, people in an area with reduced demand will be forced to leave their homes and move to distant places with potentially quite different values, habits, and climates. The real cost of migration, as well as psychic costs of adjusting to a new environment, will then be involved. The payments adjustment itself may be at best a secular and perhaps irreversible process, even if labor mobility can be relied on as an adjusting mechanism. Thus, it seems plain, without a lengthy discussion, that labor mobility is an inadequate substitute for more conventional payments adjustment instruments—demand management and exchange rate variation.

THE SHARE OF TRADABLES IN PRODUCTION

McKinnon (1963a) proposes that a high degree of openness be used as a criterion for an optimum currency area. (Externally, such an area is viewed as closed.) An "open economy" is defined by McKinnon in terms of the proportion of the production of "tradables" (both export-

² Lanyi (1969) emphasizes the barriers to interindustrial labor mobility in actual economies. Dunn (1971) stresses the heavy costs of migration and suggests that it is more humane and efficient to create demand where people are rather than to force them to move to jobs. Corden (1972) elaborates further that (1) the costs of movement would be inflicted on marginal workers to the benefit of workers who remained in employment and (2) if real wages differed among countries because of differences in union aggressiveness, migration would cause monopolistic distortions of resource allocation.

ables and importables) in the total output of that economy.³ He assumes that the output of an economy is divided into tradables and non-tradables.

In the McKinnon argument, two aspects of a small open economy lead to difficulties in using flexible exchange rates, although McKinnon does not present them in clear-cut form. These are related to the management of demand and money illusion in wage determination.

First, McKinnon emphasizes that, when an open economy (as compared with a closed economy) employs flexible exchange rates to correct external deficit, it is likely to suffer from greater price instability. When exchange rate variation is used, the domestic currency prices of tradables would change in more or less equal proportions, since the economy is open and exposed to highly elastic foreign demand for and supply of goods. When such an economy depreciates or appreciates its currency, the overall price index, involving tradables as well as non-tradables, would fluctuate more than in a relatively closed economy, thereby reducing the liquidity value of its currency. If stabilization of the overall price index were rigidly pursued, a depreciation and the resulting price rise in tradables would need to be offset by a decline in the price of nontradables, requiring a severe contraction in demand and unemployment. Thus, a number of open areas trading extensively with one another would find it beneficial to form a relatively closed common currency area.

In contrast, in an economy where the production of nontradables is substantially larger than the production of tradables, the degree of unemployment required to maintain the overall price stability is less in face of a depreciation.

It is noteworthy that, although this conclusion is derived from the model of tradables and nontradables, the latter of which display price flexibility, a similar conclusion results from application of the foreign trade multiplier analysis with constant prices. In the trade multiplier analysis, the openness is measured by the percentage of imports to national income—that is, the percentage of trade, not tradables—and the more open the economy is, the less severe the variation in income and employment becomes to correct a given payments disequilibrium under a fixed exchange rate. In other words, in a relatively open econ-

³ An "open economy" can be defined in a variety of ways, and the different definitions must be distinguished from one another. High factor mobility (Mundell) and high marginal propensity to import are factors leading to alternative definitions of openness.

omy, the use of exchange rate variation becomes less efficient than in a relatively closed economy. The benefit of a fixed exchange rate in a relatively open economy is identified with greater price stability in the flexible price model and with less income variation in the trade multiplier model with constant prices (that is, fixed exchange rates are better). Conceptually, this difference must not be overlooked. A country like the United States would be a very open economy in the sense that many of the goods produced are tradables; the economy nevertheless has a relatively small foreign trade sector. In practice, however, it seems likely that a country with a high production of tradables would have a large foreign trade sector as well.

The second aspect referred to by McKinnon as relevant in an open economy is the probable absence of money illusion. In a small area, where the proportion of imports in consumption is large, the real income effect of exchange rate variations would be so obvious to the inhabitants that they would not accept wage changes tied to a price index excluding imports. If this were the case (absence of money illusion), then the exchange rate variation would become an ineffective instrument in correcting external imbalance. In the extreme case of a totally open economy, McKinnon emphasizes, the exchange rate variations would sooner or later cause exactly equal variations in costs, depriving the flexible exchange of all its corrective functions.

There is a third (and generally neglected) argument for the appropriateness of employing fixed exchange rates in an open economy.⁴ When measuring the effects of an exchange rate change, the size and openness of the economy cannot be disregarded because a large foreign trade sector means the lack of domestic substitutes. In an economy where most of the goods consumed are imported, the responsiveness of the total amount of imports to an exchange rate change will be small, and therefore the exchange rate change required for effecting a given balance of payments improvement will have to be much larger than

⁴ See Orcutt (1955) and McKinnon (1963b). The "effectiveness" of the exchange rate depreciation can be measured by the ratio of the percentage change in the trade surplus to a 1 per cent change in the exchange rate. It is true that a small country closely approximates a perfectly competitive firm in a large market, and therefore enjoys a high price elasticity of demand for its exports in foreign markets (although this depends on what the products are). However, the relevant price elasticities here are total elasticities as opposed to individual elasticities. The point made by Orcutt and McKinnon is that, given the individual elasticities, increased openness is likely to decrease the effectiveness of exchange rate variations, if openness means a large foreign trade sector. This is an important but neglected aspect in the discussion of various elasticities in foreign trade, and it still awaits analytical clarification. See also a comment on Orcutt by Weisbrod (1956).

in an economy having a small foreign trade sector. This elasticity aspect makes flexible exchange rates a fairly efficient instrument for adjusting payments imbalance between two large areas connected by a small foreign trade sector.⁵

There are two implicit but crucial assumptions in the McKinnon theory: (1) the principal need for payments adjustments arises from microeconomic changes in demand and supply conditions, and (2) price stability prevails in the rest of the world. If the international economy itself is unstable, McKinnon's conclusion would have to be completely reversed, because external instability would be directly propagated to the domestic economy through fixed exchange.

Although this point was recognized shortly after the publication of McKinnon's contribution, Corden (1972) was the first to argue that the McKinnon criterion was not a general one. He argues that the openness criterion applies only to microeconomic demand changes that take place at home and does not apply to macroeconomic disturbances that occur abroad. If the latter is the primary cause of the payments disequilibrium, then the economy must be insulated by flexible exchange rates.

Furthermore, Corden points out that even when McKinnon's stability assumption regarding the external environment holds, one cannot say that there is a positive gain in employing fixed exchange rates; it only means that the more open the economy is, the less costly it is to maintain fixed exchange rates in terms of the required increase in the unemployment rate to effect the same degree of payments balance improvement as in the closed economies.⁶

Because of these weaknesses, the McKinnon criterion requires cautious appraisal in the practical context. It is an attractive criterion if the external environment can be assumed to be more stable than the small open economy in question. The case for the openness criterion may be strengthened if small open countries tend to be financially "undisciplined," in which case the self-imposition of fixed exchange rates is supposed to lead to sounder economic management, or if they

⁵ McKinnon (1973) examines his thesis that a currency union must follow trading relations in light of recent changes in world trade flows. While in 1973 he envisages two major currency areas—the dollar bloc and the European bloc—in a 1963 article he found that recent trade data revealed that within both North America and Europe the degree of integration had been increasing (1963b). He wants to leave open the question of whether Europe should have a currency independent of the U. S. dollar.

⁶ Price stability may be an important positive benefit of fixed exchange rates in some economies.

belong to an extreme category of the open economy where money illusion does not operate at all and the instrument of exchange rate changes becomes irrelevant. However, the assumption of the stable world economy seems to be getting less and less valid in the 1970s than in the 1950s and early 1960s, as can be seen in recent movements in exchange rates and commodity prices. If a small country is diversified in its trade with several large countries and if the exchange rates among the large countries are sharply fluctuating, the small country's peg on any single currency would increase, rather than eliminate, disturbances from the rest of the world to the domestic economy.

PRODUCT DIVERSIFICATION

Kenen (1969) argues that a low degree of product diversification is a good reason for forming an independent currency area. His argument is again based on the assumption that the principal cause of the payments imbalance is the microeconomic demand disturbances. A country that produces a wide variety of products would also be exporting a large number of different products. If there is a drop in the demand for one of the exports, unemployment would not rise so sharply as it would in a less diversified national economy under a fixed exchange rate system, simply because in a well-diversified economy each industry subject to an external shock provides only a fraction of total employment and hence the effect becomes less. Kenen also shows that when an exchange rate change is used to maintain employment at the level before the disturbance, in the event of a deterioration in the terms of trade or a downward shift in the foreign demand curve, it must be greater in a less-diversified than in a well-diversified economy. Moreover, substantial exchange rate variations would not be necessary in a diversified economy very frequently because of the averaging of the external shocks; the variety of shocks on different export products would keep export earnings rather stable. On these grounds, Kenen concludes that nations having high product diversification can tolerate fixed exchange, while less-diversified national economies should be independent (optimum) currency areas under flexible exchange rates.

Applied to the real world, the criterion of product diversification suggests that small areas with less diversification are optimum currency areas. Thus, Denmark, Iceland, Singapore, and so on, could profitably maintain flexible exchange rates. The criterion also suggests that California and Virginia should be independent currency areas because they are less diversified than the United States, although Kenen would not recommend a flexible exchange rate for regions within a country.

Flanders (1969) quibbles about the term “diversified” in Kenen’s analysis. In his formal model, Kenen uses this term only to mean an economy having an import-competing industry and not an economy exporting many products. He compares two economies exporting and importing only one good, but he should have compared two economies in which one had two export industries and the other only one. But the basic contention of the Kenen proposition seems to be so simple and obvious that it may be pointless to construct an analytical model to prove it formally.

Incidentally, in his formal model, Kenen proves the propositions stated above under very special assumptions. In particular, he assumes implicitly that the labor supply is infinitely elastic with respect to a given nominal wage rate, and that the international price of the import good and the nominal wage rate change at the same rate. If these assumptions are removed, it becomes impossible to prove the intended propositions, at least formally. It seems necessary to accept Kenen’s discussion on verbal and intuitive grounds.

It is easy to see that the low-diversification criterion comes to a conclusion that is the exact opposite of the conclusion arrived at through the openness criterion. McKinnon (1969) comments that the more diversified an economy, the larger it would be and the smaller would be the foreign trade sector. If this economy is subject to fixed exchange rates, as Kenen recommends, then fiscal-monetary policy must be used to correct external imbalance—the undesirable “tail-wagging-the-dog” situation.

These contradicting conclusions arise from different assumptions concerning the principal source of the payments disequilibrium. To simplify, McKinnon is concerned with internal shocks, while Kenen is interested in external shocks to export products. When instability is assumed to exist in the rest of the world, it is natural to have a flexible exchange rate to insulate the economy from outside disturbances. Such a difference in prescription indicates the importance of the basic assumptions.

DEGREE OF FINANCIAL INTEGRATION

The criteria discussed above refer to real, economic characteristics of the economy. Ingram (1969) thinks that analyses such as those presented by Mundell, McKinnon, and Kenen leave little room for money; in their models, prices are expressed in real terms of trade and all external adjustment occurs on current account, eliminating the inter-

esting issues altogether. Ingram claims that it is necessary to look at financial, not real, characteristics of economies in order to determine the optimum size of a currency area.

Ingram (1962a, 1962b, and 1973) and Scitovsky (1958 and 1967) are inclined to employ a high degree of international financial integration not only of short-term but also of long-term securities as a criterion for an optimum currency area, although their analyses are not directly addressed to the present problem. They expounded the point that if there is a high degree of integration in the financial capital market in an area, then there will be no need for flexible exchange rates.

Ingram (1962a) states that if international financial integration is insufficient, the difference in the interest rate structure will be more apparent in longer-term rates, because foreign purchases and sales of securities tend to be concentrated in short-term ones in which foreign exchange risk can be covered in the forward market. The absence of free transactions in long-term securities among nations, Ingram asserts, will be a source of balance of payments instability because foreign holdings of short-term claims tend to be liquidated if and when the interest rate differential (allowing for cost of forward exchange cover) declines. Thus, the most important aspect of the financial integration criterion relates to long-term securities; mere integration of markets for short-term funds is not considered sufficient as a condition for a common currency area.

Ingram does recognize the possibility of speculative capital flows when exchange rates are fixed and financial markets are artificially separated among countries by direct controls. What he really means is that under a high degree of financial integration, the need for exchange rate changes would be eliminated because only fractional changes in interest rates would evoke sufficient equilibrating capital movements across national frontiers. These equilibrating flows would come into being when each government agreed to a rigid, permanent link among currencies and eliminated all legal restrictions on international payments in order to achieve the desired degree of financial integration. Ingram argues that intended separation of financial markets, which some governments still seem to pursue primarily by direct controls—and which, nevertheless, is frequently frustrated by evasions—would restrain a vast amount of potentially equilibrating capital movements.

The Ingram criterion creates several difficulties. First, Fleming's comment on capital mobility, already reviewed, can be invoked here again; even if Ingram's presuppositions are satisfied, there may still be dis-

equilibrating capital flows. Second, in the present world as it stands, there is no large stock of internationally acceptable financial assets. Residents of payments surplus countries are basically not willing to lend extensively to deficit countries.

Furthermore, Ingram's analysis is attacked by those who make a conceptual distinction between financing and adjustment. Tower and Willett (1970) comment briefly that the mechanism suggested by Ingram and Scitovsky would primarily be financing, rather than correcting or adjusting, payments imbalance. Corden (1972) expresses more explicitly a negative view on the Ingram approach, arguing that it is merely a short-run stopgap measure, much like the Mundellian argument (1962) for manipulating the fiscal-monetary policy mix for the simultaneous realization of internal and external balance.⁷

It is always useful to make a distinction between financing and adjustment. A payments deficit may be financed by the transfer of financial claims, thereby allowing real adjustment (reduction in expenditure) to be spread over a long period. It is not clear, however, whether the ease of financing always implies a smaller total adjustment cost for an area in which a payments deficit has occurred. This is essentially an empirical question that should be examined.

These weaknesses of the Ingram argument seem to be sufficient for rejecting it as a tenable criterion for resolving the problem of optimum currency areas.

SIMILARITY IN RATES OF INFLATION

The criterion of similar inflation rates is explicitly put forward by Haberler (1970) and Fleming (1971). They claim that this factor is at least as important as any of the factors that have been discussed, and yet it has occupied a much less prominent place in the literature. This criterion involves a change in the basic assumption regarding the principal source of the payments imbalance.

The argument that attention should be focused on the inflation rate is that the payments imbalance is most often likely to be the result of divergent trends in national inflation rates due to structural developments, differences in trade union aggressiveness, or differences in national monetary policies. This criterion, therefore, diverts attention from microeconomic disturbances in demand and supply conditions and focuses it on macroeconomic phenomena.

⁷ In reconciling these views, Salant (1973) offers an observation that capital flows promote adjustment of current accounts to the extent that wealth influences expenditure; more borrowing would mean less expenditure.

The usefulness of the inflation rate criterion thus depends on the significance of differential rates of inflation and productivity growth as a source of payments disturbances. It seems that a definite conclusion has not yet been reached as to whether they are indeed *the most important* or *the most frequent* factors in causing payments balance problems. Interestingly, there seems to be a discernible tendency, in general, for the academically minded to become interested in micro-economic disturbances (hence, their interest in exchange flexibility), while the practically minded economists and government officials attribute the major cause of payments imbalance to inflation and inadequate demand management policy.

The usefulness of the inflation rate criterion also depends on the degree of intractability of divergent cost and inflation trends among nations. If, as is commonly thought, countries have their own inflation-employment trade-offs, then it is costly, in terms of deviations from the nationally desired positions on the Phillips curves, to reduce the divergent trends. This proposition, however, is still unproven, and more empirical investigations are awaited before a firmer judgment can be made. If differential rates of inflation are not an intractable problem, as Parkin (1972) asserts, then the fact that different countries happen to have had differential rates of inflation in the past would not be a sufficient reason for attaching the greatest importance to this criterion.

DEGREE OF POLICY INTEGRATION

Finally, Haberler (1970), Ingram (1969), and Tower and Willett (1970) stress that it is not so much economic characteristics as the similarity of policy attitudes of member countries that is relevant in making a group of countries a successful currency area, although they were vague about the content of "policy integration."

Tower and Willett (1970, p. 411) state:

Perhaps of primary importance for a successful currency area with a less than perfect internal-adjustment mechanism is that there be a reasonable degree of compatibility between the member countries' attitudes toward growth of inflation and unemployment and their abilities to "trade off" between these objectives. A nation with a low tolerance for unemployment, . . . and price pressures from concentrated industries, would make a poor partner for a country with a low tolerance of inflation and high productivity growth, making for a very favorable "Phillips Curve."

This policy integration criterion is similar in approach to the inflation rate criterion, but it involves diverse elements and is not very homogeneous.

In discussing this policy integration criterion, Fleming (1971) points out that some types of policy coordination do not contribute to the mitigation of payments disequilibria. He states that, in the field of fiscal policy, regions within countries that are economically depressed would be helpful in correcting payments imbalance only to the extent that prosperous countries or regions of the area are in a position of payments surplus and depressed countries or regions are in a position of deficit. Only then a unified fiscal policy (a transfer from prosperous to depressed regions) can contribute to mitigate or eliminate payments imbalance.

Fleming also expresses doubt about what effects advance in policy integration could have in the field of monetary policy. For example, centralization of banking through the establishment of a supranational central bank for the area would encourage equalization of short-term interest rates and mobility of capital throughout the area; such centralization is not always helpful in mitigating payments disequilibria among member countries of a monetary union.

In assessing the significance of policy integration, it may be recalled that the question of temporal ordering of exchange rate fixation and policy coordination remains unresolved. The Werner Report of the Commission of the European Communities (1970) clearly acknowledges that nations must ultimately surrender their monetary sovereignty to establish a complete monetary union in Europe. Serious policy coordination among sovereign nations, however, demands a great deal of political will and determination and becomes a domestic issue which may be highly sensitive politically. Thus, commitments could be made to rigid exchange rates (prematurely) in the European Communities but not to a program to transfer national policy autonomy increasingly to a supranational organ.

Recent events in Europe—particularly the rupture of the European currency “snake” may be taken as endorsing the view of the so-called economists that policy coordination ought to be arranged prior to exchange rate fixation.⁸ The question, however, is similar to that of balanced versus unbalanced growth in development strategy of a developing nation. A compromise between the integrationist approach and the exchange rate approach to monetary unification may always be possible

⁸ In the European Communities the controversy between the economists and the monetarists has been well publicized. The former assert that a common currency in Europe should be established only after economic conditions for it mature, whereas the latter favor the precedence of monetary integration in the belief that it will force member countries to coordinate policies. Compromises between these two schools of thought are proposed by Cohen (1963), Magnifico and Williamson (1972), and Balassa (1972).

and necessary, given limitations imposed by political conditions in member countries.

II. An Alternative Approach

SOME OBSERVATIONS ON THE TRADITIONAL APPROACH

In what has been termed above the traditional approach to the question of optimum currency areas, the contributors to the literature point out the cases where exchange rate variation is either ineffective (the absence of money illusion, unalterable real wages) or unnecessary even though perhaps effective (high factor mobility or product diversification and similarity in inflation rates), as an argument for having a common currency. In cases where exchange rate variation is effective in correcting payments imbalances, they try to evaluate the costs of payments adjustment involved in the exchange rate depreciation and deflation with fixed exchange rates in terms of the objectives of price stability and full employment or their variants.⁹ However, all the analyses have very little to say about the positive benefits of a common currency.

Moreover, advocates of different criteria have not sufficiently emphasized the importance of their assumptions with respect to the cause, magnitude, duration, and frequency of payments imbalances, nor have they included them in their models. Recent experience in exchange rate problems supports the importance of differential rates of inflation, and the discord of national demand management policies behind them (emphasized particularly by Haberler), as a major source of payments imbalances. These differential rates of inflation might be traced to the differences in social preference functions (sociological forces) affecting wage increases, or to the law of dynamic increasing returns as emphasized by Kaldor.¹⁰ In the judgment of the author, the pervasiveness of these forces

⁹ A formalization of this idea can be found in Kreinin and Heller (1974). They made the (marginal) cost of depreciation a terms-of-trade deterioration and compared it with the (marginal) cost of adjustment under a fixed exchange rate regime, that is, the amount of expenditure reduction required to improve the payments balance by one unit (the inverse of the marginal propensity to import). The boundaries of optimum currency areas can be drawn when these two costs balance. Sellekaerts and Sellekaerts (1973) extended the idea of the marginal cost of adjustment to the case of compensatory capital inflow.

¹⁰ In the presence of increasing returns, the rates of growth of productivity will be higher, the higher the rates of growth of output. It has been generally observed that the difference between rates of increase in productivity and money wages is smaller in rapidly growing regions than in lagging regions, and for this reason these regions acquire a cumulative competitive advantage over slow-growth regions. See Kaldor (1970, 1971, and 1973). Kaldor (1971, p. 78) claims that,

as a source of payments imbalances is so overwhelming in practice that it seems that theories based on microeconomic disturbances have to be rejected as a guide for practical policy. The microdisturbances in the payments imbalances are likely to be only temporary and reversible (in which case there would be no need for exchange rate variation), while divergent wage and price trends would derive from deep-seated (and persistent) social forces.

The deficiency of the contributions can also be found in the lack of explicit and precise models, although the costs of payments adjustment have emerged as the central concept in the choice between alternative exchange rate regimes. Adjustment costs, of course, depend on the model used. Some contributors use the static Keynesian model and some use the tradables and nontradables model or the Phillips curve type model; the implications of these models for adjustment under fixed and flexible exchange rates vary.¹¹ In the static Keynesian model, the adjustment cost under a fixed exchange rate can simply and convincingly be expressed as the reduction in real absorption, but the adjustment cost arising from the use of a flexible exchange rate can be expressed in a variety of ways—price instability, deterioration of the terms of trade, and the temporary costs of the reallocation of resources in accordance with the new price system.¹² At any rate, there seems to be a need for further elaboration and research.

The fundamental objection to all the contributions to the literature perhaps lies in the lack of the idea of balancing the *positive* gains of fixed exchange rates or a common currency against the losses. A country participating in a currency union renounces a very important instrument of economic policy; it is making an irreversible commitment to fixed exchange rates. With this abandonment of exchange rate policy would go a host of other domestic policy instruments, particularly independent monetary policy. What would be the gains for this country in exchange for this sacrifice?

If *ceteris paribus* conditions could be assumed to be valid, it is necessarily true that national welfare cannot be reduced by giving the authori-

because of the presence of regional problems, the objective of a full monetary union is unattainable without a political union, which presupposes not just fiscal harmonization but fiscal integration.

¹¹ The cost of adjustment under a fixed exchange rate in a Phillips curve economy may be measured by unemployment in excess of the preferred point on the Phillips curve.

¹² See the distinction between the "transitional cost" and the "continuing cost" by Cohen (1966).

ties the instrument of exchange rate variation.¹³ Governments retain an extra degree of freedom by keeping the exchange rate at hand as a policy instrument, even though this policy may not be activated. In practice, however, several benefits accrue to a country joining a currency union and renouncing an independent exchange rate policy, such as the greater extent of liquidity value of money and the elimination of exchange speculation and of the cost of holding external reserves. The declared trade-off in the original Mundell contribution was indeed between such positive benefits and the losses caused by abandoning the freedom of exchange rate policy; but Mundell did not carry out the full analysis of the multiplicity of costs and benefits. To be sure, it is difficult to evaluate these costs and benefits and to weigh them against one another. But such a multidimensional approach, taking a wider range of benefits of a common currency into account, seems to be not only more comprehensive than any single "criterion" approach that was reviewed but also superior to it. It is time to abandon the conventional search for a single criterion on which the desirable extent of currency domain can supposedly be identified.¹⁴

THE BENEFITS

Let us first consider the benefits of a common currency. It may be recalled that the theories of the traditional approach are preoccupied with achieving the standard macroeconomic policy targets. This means that they implicitly assume a very simple type of social preference function—a fixed target function. Without going into the problem of strengths and weaknesses of this approach, which has been extensively discussed in the literature, suffice it to say that it would be fruitful to bring in some dynamic elements, even on an intuitive basis, since balance of payments adjustments, like most economic adjustments, never take place instantaneously but only over time.

Currency unification—and monetary integration more broadly—is a dynamic, evolutionary process. Thus, it would be a mistake to pass judgment on the desirability of a given country joining a currency union on the basis of static gains and losses pertaining only to price stability, employment, and payments equilibrium. As discussed above, there would be other, more specific, gains and losses of a more dynamic nature,

¹³ See the discussion of the welfare cost of maintaining a fixed exchange rate by Hause (1966) and Johnson (1966).

¹⁴ Lamfalussy (1974) can be cited as an example of one who abandons the single criterion approach. Williamson (1974) applauded Lamfalussy for this but questioned his potential overestimate of the benefit of intra-EEC trade and the neglect of fiscal integration that would be prompted by monetary integration.

which are subject to constant change in the course of progressive monetary unification. For a comprehensive judgment, all these pros and cons ought to be taken into account, even though they are extremely difficult to weigh. (In a broader sense, the issue definitely falls into the realm of welfare economics, and the ultimate decision inevitably rests on strong political value judgments.) In other words, it would be necessary to discuss not only the currency union as a complete object, but also the process of currency unification. In terms of the progression of monetary integration, several important gains of a common currency can be enumerated.

First comes the familiar argument of the value of money in an open economy.¹⁵ Money, if it is to be money, has to serve as a universal measure of value, a universally accepted medium of exchange, and a store of value. A common currency as the medium of exchange eliminates the costs of money conversion and forward cover required in a flexible exchange system. A currency of a large area would also be a good store of value; ignoring inflation, such a currency can command a wide selection of commodities that are free from price changes due to exchange rate changes. In these respects, therefore, a common currency is conducive to allocative efficiency and integration of the economy.¹⁶ Indeed, on this basis alone, the optimum currency area must be the world; the existence of numerous small national currencies is viewed as a "barbarism"—a peculiar assertion of nationalism. However, one would hesitate to give great weight to this argument in the practical context. Of course, some minimum fixed exchange rate area is necessary if money is to exist at all. The point is that the small countries (such as Denmark, Switzerland, Singapore) are not so small that they have to peg to another currency in order to secure the monetary value of their currencies.

Secondly, speculative capital flows will be completely eliminated, thereby relieving the authorities of frustration in their monetary control.¹⁷ In a small country, the exchange market would be very small in size, allowing a small number of speculators to collude and influence the market. In addition, Fleming (1971) thinks that changes in exchange rates between members of the area and the outer world will probably become less frequent and smaller. For example, no change in the unified exchange rate may occur in situations where, in the absence of

¹⁵ This aspect was particularly emphasized by Mundell (1961) and Tower and Willett (1970).

¹⁶ The Common Agricultural Policy in the European Communities can be viewed as such an attempt.

¹⁷ See Mundell (1973b).

exchange unification, one member might have revalued and another devalued its currency.

This conclusion, of course, requires qualification to the extent that the fixity of exchange rates may not be perfectly certain, depending on the particular evolutionary stage of a currency union. When an exchange rate change within the currency union threatens, and yet the member involved is discouraged from making a prompt decision due probably to overall reluctance of the union to disrupt the unified rate, speculative capital flows may become larger than they otherwise would have been. Since, in practice, irrevocably fixed exchange rates are hard to imagine among nations, unless they achieve political unification, one would again be reluctant to give much weight to the benefit of no speculation.

Another qualification is that the question of whether it is best to fight speculation or its effects is still an unsettled matter. Even if exchange rates among participants can be fixed once and for all to eliminate speculation in the area, the alternative policy of flexible exchange rates may be more effective in retaining government control of monetary policy (see Balassa, 1973b).

Thirdly, there is the benefit of saving on exchange reserves. The importance of this benefit is doubtful in the early stage of a currency union, where the members are still not fully convinced of the future course of events, but it will become increasingly significant as cooperation among the member countries advances. Ultimately, the members would be completely liberated from having external reserves for transactions internal to the union, just as would regions within a country. How much reserve saving can be realized in the interim stages of currency unification is a policy question.

Kafka (1969) pays some attention to this problem and recognizes two aspects of it. First, it is obvious that participants in a currency union can economize on reserves, as far as intra-area trade is concerned, by granting each other credits. Secondly, he states, a group of countries can economize on reserve needs vis-à-vis third countries if such needs are mutually offsetting and if the group pools its reserves. Though he is rather skeptical as to the benefit of a currency union with respect to reserve saving if the union consists only of developing countries, since their mutual trade is limited, Kafka states that there is no harm in attempting to gain the benefit. He also refers to the possibility of reserve saving through realization of direct payments channeled throughout a union rather than through the great financial centers.

Kafka (1973) takes up the subject again in the context of Latin American monetary integration. He considers that Latin America is not an optimum currency area either now or for the foreseeable future, but concludes that some reserve-pooling arrangement on a limited scale may be desirable. He is not in favor of large credit facilities within Latin America because they are likely to result in the expansion of regional protection (Kafka thinks that the necessity to ensure liquidation of the clearing credit is likely to induce countries to employ more exchange restrictions), and because they may be useless if payments difficulties of the member countries arise vis-à-vis nonmember countries.

Most recently, Mundell (1973a) attempts to specify more explicitly the benefits of a common currency. To him a common currency has as many as six benefits—reserve saving, a reduction in the cost of financial management, risk pooling, intermediation, information saving, and innovation. Of these, great importance is attached to reserve saving, which can be realized by either swap arrangements or a reserve pool.¹⁸

Fourthly, further arguments for fixed exchange rates appear in the literature. The benefit of risk pooling is emphasized by Mundell (1973a), and the proposition that a fixed exchange rate leads to less costly payments adjustment and efficiency in resource allocation and money holding is presented by Laffer (1973). Intuitively, Mundell's argument, as well as one of Laffer's is based on the idea of spreading economic distress over space and time.¹⁹ Their economic models and arguments, however, are still vague and await further research. The author believes that Mundell failed in his main argument to define unambiguously and to prove the existence of the benefit of risk pooling. Other benefits of a common currency, as noted by Mundell, also remain vague and intuitive.

Fifthly, and probably much more important potentially than the foregoing, monetary integration is likely to accelerate fiscal integration. Williamson (1974) states:

In the first instance one might expect monetary union to accentuate regional problems, on account of demonstration effects tending to raise wages in regions where productivity was relatively low. Because unemployment is more visible than poverty, and because a regional policy to remedy unemployment has a minimal cost to the community as a whole, one can expect the intensification of regional problems to stimulate political pressures for fiscal integration.

However, although this benefit may be potentially of major importance, it may turn out to remain unrealizable in the foreseeable future

¹⁸ Salant (1973) made a more careful analysis of possible saving in external reserves in the light of recent experience in the European Communities.

¹⁹ See a criticism on Mundell by Balassa (1973a).

in any monetary union—just as the attempt to fix exchange rates in the European Communities did not significantly accelerate integration in monetary and economic policy. Unlike Williamson's expectation of "minimal cost," it seems that fiscal integration would require the creation of a supranational fiscal authority—something very close to supranational government—or at least a large-scale regional policy of a monetary union. This is precisely the area in which the European Communities have made hardly any progress. Whether such extensive fiscal integration materializes is indeed a political issue that requires concessions on the part of prosperous member countries.

THE COSTS

From the point of view of the individual nation, either a common currency or an irrevocably fixed exchange rate involves serious economic losses—perhaps more serious in comparison with the benefits enumerated above.

First, the obvious main implication of a common currency is the loss of autonomy in monetary policy (see Lutz, 1972). Participation in a currency union means the surrender of the power to alter national money supply, leads to the creation of the integrated capital market, and eventually involves a supranational central bank. This loss of national control over the exchange rate may quite legitimately be feared and regarded as dangerous by the national government; as long as differences in national wage, price, and productivity trends persist, there is a danger that the chronic surplus country will aggravate domestic inflation and the deficit country will suffer from further depression and unemployment, depending on the nature of the coordinated overall monetary policy of the union. The jointly controlled monetary policy would be designed to serve the overall or majority interests of the member countries, and it is not necessarily beneficial to particular regions or nations. As Mundell (1962) and Fleming (1962) show in their classic contributions to the literature, national monetary policy becomes incapable of affecting employment.²⁰ If one country tries monetary expansion, a downward pressure on the interest rate in that country would cause a capital outflow and a deterioration in the payments balance, which would absorb domestic money. This process continues until the accumulated deficit restores the stock of domestic money to its original level. Thus, all real variables would remain unchanged.

²⁰ Sohmen (1969b) points out that the asserted inability of monetary policy under fixed exchange to influence employment still holds under assumptions less austere than those made by Mundell (1962) and Fleming (1962). McKinnon and Oates (1966) discuss policy implications of international economic integration.

Secondly, there is an implication for national fiscal policy. It is commonly thought that fiscal policy can be left for the national government as an effective instrument for influencing domestic employment; expansion of government expenditure or tax reduction would be relied on to increase income and employment.

It is, however, not clear how free the national fiscal policy can be in a currency union. It is quite conceivable that a currency union imposes coordination of goals and policies—including fiscal policy—on the participating countries so that the union can maintain its joint external payments equilibrium. How compelling is this force of coordination can be examined by considering a model of at least three countries, such as the one constructed by Arndt (1973). His discussion of the policy system of a currency area is as follows. Suppose that two countries form a currency area, leaving a third on the outside. (It is assumed that capital mobility is perfect within the area but is imperfect between that area and the rest of the world.) In such a world, a number of peculiarities arise which do not appear in a single country or in a two-country world. For example, in a two-country model with imperfectly mobile capital, each country is provided with two instruments (one fiscal, the other monetary), making four instruments altogether. One of these is redundant, however, because one country's surplus exactly equals the other's deficit. If, however, there are three countries, the symmetry in the balance of payments no longer exists; one country's balance of payments is a composite of its trade with the other as well as with the third country. Thus, insofar as the two countries in the currency union continue to worry about both intraregional and external trade balance, the number of instruments becomes less than the number of targets. Arndt suggests that this difficulty could be alleviated by coordinating fiscal policy *in addition to* the obvious need for coordinating monetary policy.²¹

Johnson (1971) considers the implication of a centrally created common currency for freedom in fiscal policy. He stresses the constraint

²¹ Arndt assumes that the relationship between the currency area and the third country is that of a fixed exchange rate also. This is not an appropriate formulation, however. More appropriately, a situation in which two countries jointly float their currencies vis-à-vis the third country's currency must be considered. This constitutes the addition of one more instrument, providing the area with the four instruments. Arndt's concern with the intraregional balance is valid only for a pseudo-currency union.

Swoboda (1973) addresses himself to essentially the same problem, but in the context of a two-country model. He also discusses the problem of a shortage in the number of instruments when the level of expenditure is the only policy available for each country.

that the governments can only finance deficits by borrowing in the international capital market on competitive terms, and not by the creation of money in their favor, just as local governments can within the country.

Thirdly, assuming that each country has its own particular Phillips curve, a common currency area may imply "the worsening of the unemployment-inflation relationships." Fleming (1971) identifies this case as a major difficulty arising from monetary integration; he asserts that the fixation of exchange rates will increase the amount of unemployment required to hold inflation at any given rate and will increase the rate of inflation corresponding to any given level of unemployment in the area as a whole. The basic reason for this is the curvilinear relationship between unemployment and inflation involved in the Phillips curves.

Fleming also considers a situation in which policy coordination is absent and expenditure policy is the only available instrument in the country. He illustrates this point by an example of two countries having identical economies and Phillips curves. The two countries are assumed to be in external equilibrium and at the same point on the curve. If the payments equilibrium is disturbed, one country has to deflate and the other has to inflate. In considering a situation in which the level of employment as a whole is constant—that is, the increase in unemployment in the deficit country is exactly matched by the decrease in unemployment in the surplus country—the resulting average rate of inflation must be higher than the one that prevailed before the disturbance, because the increase in inflation in the surplus country is greater than the decrease in inflation in the deficit country.²² (Fleming, however, does not neglect the possibility of reduced average inflation. If the deficit country's initial position on the Phillips curve is higher than that of the surplus country, the average rate of inflation will decline as a result of any payments adjustment that keeps the total level of employment in the two countries unchanged.)

The argument of the greater possibility of unemployment can still be based on a likely asymmetry in adjustment. The low-inflation surplus country in the area would perhaps dominate the scene and would force the others to adjust because the former normally feels less pressure to adjust. If this is the case, although the average rate of inflation would be lowered, there would be a rise in unemployment in the area as a whole. Indeed, this is a more likely outcome than a situation envisaged

²² This argument, based on the recognition that labor is not mobile enough and that coordination of fiscal policy is extremely difficult in Europe, is the major pillar of Fleming's case against the kind of currency unification presently aimed at in Europe.

by Fleming in which the two countries seem to share the burden of adjustment more or less equally. In this sense, then, there would be a worsening of the unemployment-inflation relationship as a result of currency unification.

Parkin (1972) presents an almost opposite view. Addressing the same problem, he concludes that a monetary union has a favorable effect on the inflation rate of a member country. Citing the study of Laidler (1973) on the recent experience of the United Kingdom in inflation and devaluations, he propounds the monetarist view that in an open economy with a fixed exchange rate the inflation rate cannot deviate much from that prevailing in the rest of the world; excessive expenditure relative to income would spill over into a trade deficit, exerting competitive pressures on wages and prices to keep them in line with those in a monetary union. If this view is correct, the forces of international competition could be relied on for both internal and external balance, and serious deflations or devaluations would not be needed. On the Parkin paper, there is a rebuttal by Hirsch (1972b). Resolution of this controversy would require more input of genuine econometric studies.

A fourth aspect of cost associated with a common currency area is the possible deterioration of regional economies. Because of the relative ease of capital movements as compared with international labor migration, the theory is widely held that monetary integration is likely to result in the acceleration of economic distress and in stagnation of certain regions or countries.²³ Whatever the explanation may be, it has been a fact that differences in the rates of increase of money wage rates in the various regions, or different countries in Europe, tend to be smaller than the differences in the growth rates of productivity in those regions. This may indeed be due to the phenomenon of dynamic increasing returns, or what Kaldor terms "the Verdoorn Law." According to this law, the fast-growth regions attract capital because of their

²³ See Johnson (1971), Hirsch (1972a and 1972b), and Kaldor (1970 and 1971). Williamson (1973) has discussed alternative explanations of the persistence of the regional problems in a monetary union. To the extent that each region of the monetary union has its own Phillips curve, regional unemployment will be the more severe, the greater the degree of monetary integration in the union. In contrast, if regionally separate Phillips curves are only a transitory phenomenon, as Parkin (1972) considers in view of the adaptation to the common rate of inflation in a monetary union, then the regional unemployment problem also becomes transitory in nature. Williamson constructed a comprehensive model which can accommodate views of those and other alternative theories, and concluded that, since such a general model can explain apparent facts, one has to avoid dogmatic judgment as to whether a monetary union adds to the severity of the regional problem.

lower unit labor costs relative to the slow-growth regions, probably accentuating the unemployment problem of the latter. In such a context, some economists seriously question whether Scotland or Ireland or Southern Italy might not have been better off if they had been able to have their own currencies and depreciate them. Even if the growth rates of productivity were equalized in the regions of a currency union, the concentration of capital in the more prosperous regions would still take place if there remain national or regional differences in wage push.

Regional or national distress in a monetary union can be expected to be compensated to the extent that the union develops a supranational fiscal transfer system. Within the nations, compensation has been offered in the form of explicit transfers to depressed regions or lower tax revenues from depressed regions. However, the success of regional policies on an international level may well be less than those on a national level.

THE QUESTION OF WEIGHING

Assessment of the benefits and costs enumerated above is very difficult. The benefits and costs, or the pros and cons, are not static and are not to a great extent commensurate; a precise quantification of these effects of a common currency in a simple measure like real income streams would presumably be impossible. The difficulty of quantification is aggravated if there are interdependencies among the benefits and costs; their magnitudes, then, cannot be measured separately. Nevertheless, these are the considerations that must influence a country in deciding whether or not to join a common currency area.

In reality, however, it may be possible to dispense with precise quantification. A country pondering over joining a monetary union would naturally regard the above-mentioned benefits to be of a somewhat abstract nature, which is in contrast with the clearly visible losses of high inflation, greater unemployment, and so on. In the decade-long debate over U. K. membership in the European Economic Community (EEC), most British economists were deeply skeptical of the benefits and feared that most of the burdens would fall on the United Kingdom, particularly on workers.²⁴

The commitment to a fixed exchange rate without an underlying agreement to surrender national policy autonomy formed what Corden

²⁴ Hirsch (1972a) suspects that the City of London may gain from European monetary integration at the expense of U.K. workers. If so, a serious question of income distribution is also involved.

(1972) called a "pseudo exchange rate union" in Europe, and simply aggravated domestic policy management. The experience of the past two years in Europe does bear out these fears about the costs of monetary integration and suggests that the benefits such as promotion of growth through trade interpenetration, acceleration of policy harmonization, and so on, have not materialized. In Europe, trade interpenetration and capital movements remain surprisingly low,²⁵ in comparison with those among states in the United States. In view of these facts and of the persistence of significant differences in policy attitudes among nations, one cannot escape the conclusion that a member of the EEC at the present stage can only incur very high costs by committing itself to schemes such as the currency "snake."

An extremely interesting question in relation to weighing pros and cons is one of whether Europe is already in a situation where parity changes are ineffective, so that pegging or varying the parity has lost significance. As has been discussed, some monetarists take the view that product markets in Europe are completely integrated and that therefore price levels are internationally determined. Devaluation, then, becomes just another form of monetary restraint. The truth is that trade interpenetration is still low in Europe. Devaluation is still a meaningful instrument not only to influence relative prices and divert resources into the trade balance when there are unemployed resources, but to cut domestic consumption through the "money illusion"—the fact that consumers are less resistant to an indirect cut in real consumption through devaluation than a direct cut through a reduction in money wages.²⁶ (The preservation of the "money illusion," of course, depends on the condition that devaluation has not become a habit.) Moreover, as Oppenheimer (1973, pp. 107-108) points out:

The point . . . is not that the individual countries of Western Europe have already ceased to be feasible currency areas, but that governments, by their indiscriminate pursuit of integration in the economic and monetary sphere, may unwittingly bring this result about; and may afterwards regret it, if they decide that EMU is not such a good idea after all.

To date, economists have suggested that it is possible to make an overwhelming case either for or against monetary integration.²⁷ In some trivial cases this may indeed be true. But when a multitude of countries

²⁵ For trade, see Kaldor (1971). The evidence is hard to find for capital flows, but the indirect evidence is the total absence of a common policy for the EEC concerning exchange restrictions and differential tax systems.

²⁶ See Hirsch (1972a and 1972b) and Oppenheimer (1973).

²⁷ See a (rather dogmatic) controversy between Parkin (1972) and Hirsch (1972b).

of substantial size or importance are involved (such as the EEC), it seems that the evaluation of the benefits and costs requires a substantial amount of both conceptual and quantitative research, as well as regular updating in response to the changing reality.

III. Further Issues

IMPLICATIONS FOR THE WORLD EXCHANGE RATE SYSTEM

The point consistently stressed in the discussion of currency areas is the finding that, even if the argument for flexible exchange rates is accepted, there still remains the problem of how the world should be divided into currency areas.

Mundell (1961) points out that the conflict between the case for a flexible exchange rate and the case for a common currency in a given region is empirical rather than theoretical. In a well-known paper, Meade (1957) argues for flexible national currencies in the member countries of the European Community, pointing out that there is not sufficient labor mobility, while Scitovsky (1958) favors a common currency because of its effect in encouraging capital mobility. Thus, both writers implicitly accept the proposition that the optimum currency area is a region defined by high factor mobility. What they dispute is whether Western Europe as a region is already sufficiently integrated to have a common currency. (Note, however, that, as proponents of the policy integration criterion show, mere integration of private economic activity is not likely to be a *sufficient* condition for having a common currency.)

Johnson (1970), for example, finds three defects in the case for fixed exchange rates: (1) lack of labor mobility over national boundaries, (2) lack of an international government to mitigate national distress, and (3) lack of an international central authority to control the overall quantity of money. But these are all empirical, rather than theoretical, objections to the adoption of fixed exchange. The important question is the *degree* of the lack of conditions that assure smooth functioning of a fixed exchange rate regime. No one would believe that the age of international integration of economic policies in Europe is close at hand; but neither would one think that permanent institutionalization of a flexible exchange rate regime in Europe is necessary or beneficial.

The working of a system consisting of a few monetary groups or blocs has been discussed within the limits of the optimum currency area approach. Suppose that small national currencies are driven out by

a more efficient producer of money, or small countries voluntarily peg their currencies to a dominant currency because of convenience, or an entirely new common currency is created. Then, how does such a system work?

Mundell (1965) offers some speculations. He identifies a major problem in the method of settling accounts between a few currency areas and discusses three possibilities: (1) the gold standard; (2) flexible exchange rates among the dominant currencies; and (3) the loosened form of the gold standard in which dominant currencies are pegged to gold, but at substantially widened gold margins. Mundell's inclination to rely on gold stems from the belief that politically distinct monetary blocs would be unwilling to hold large external reserves in the form of currencies of other blocs. One may well disagree with this, but if there is an major advance in European monetary integration in the future, this consideration will assume importance. The world monetary system designed by the optimum currency area approach is a multipolar, dual system, and although the benefits of a common currency would be more genuine within each monetary bloc than in the Bretton Woods type of system, the whole payments mechanism would become quite complex.

FIXED EXCHANGE RATES VERSUS COMMON CURRENCY

Although the above discussion has taken for granted that pegged exchange rates automatically imply full convertibility, the further implications of this have not been explored. Sohmen (1969b, p. 194) elaborates on these implications of perfect currency convertibility:

Within a currency area, it is not merely *spot* "exchange rates" that are constant over time (without limits of fluctuation, moreover). Since there is complete assurance that claims on debtors in other regions can always be repatriated at the same "exchange rate" of 1:1, no matter how long their maturities may be, all forward "rates of exchange" between regions are in effect fixed at the value 1:1 at all times. . . . The present state of affairs, or anything we can reasonably hope for in our grandchildren's lifetimes, does not have the remotest resemblance to it.

Since governments today are far from being capable of pegging the exchange rates for all maturities, he concluded that to secure the benefits of a currency area, there must be the creation of a single common currency, not the pegging of exchange rates. This, of course, entails the creation of a single central monetary authority in the currency union.

The differences between systems of a single common currency and multiple national currencies pegged to each other may be great. Johnson

(1963), Ingram (1962b), and others have emphasized that exchange rate fixation alone would not lead to the acceleration of capital mobility and competition within the area, since, in defense of the fixed rate, governments may well be obliged to resort to restrictions on transactions involving foreign exchange. The point was also made by most contributors to the literature, and in particular, Meade (1957). Thus, there is reason to be skeptical about premature fixing of exchange rates at an early stage in the EEC (see also Presley and Coffey, 1972).

The thrust of the arguments by Sohmen (1969a and 1969b) and Johnson (1963) is that serious monetary integration can only be carried out under a common currency and a central supranational monetary authority—a point to which early arguments for currency unification were somewhat oblivious. They have overemphasized their point, however. Currency unification could proceed from mere fixation of exchange rates to the establishment of various supranational authorities, to the creation of a common monetary unit, and to full coordination of economic policies. As the confidence in rates becomes firmer, the forward markets themselves would become unnecessary and eventually disappear. In view of the different conceivable stages of currency unification, it seems unnecessarily extreme to claim that a single common currency must exist from the beginning and then to say that a common currency is not feasible in practice and that therefore fixed exchange rates would lead to disaster.

Magnifico and Williamson (1972) express a preference for a more gradual approach to monetary unification in Europe. Their argument is based on scale economies of a common currency and the need for protection in the early stage of its inception (analogous to infant industry within the nation), while allowing national differences in exchange rate and economic policies to continue until the common currency is firmly established. This argument differs from that of the Werner Report in which heavy emphasis is placed on the mere convergence of existing national currencies.

CREATION OF A COMMON CURRENCY

Except for the use of gold, there are basically two ways of creating a common currency—to internationalize one of the existing national currencies or to create an entirely new unit of account.

In cases where small developing countries and an overwhelmingly large developed country are involved, it is likely that the currency of the

dominant economy becomes the currency of other countries as well. Such cases are often observed among countries that were formerly the metropolitan country and its colonies. In such instances, the small countries would peg their currencies irrevocably to the currency of the mother country but would normally retain their own currency standards. In an extreme case, however, the small country would not have locally issued currency in circulation, but would use the mother country's currency.

Johnson (1973b) examines the costs and benefits of such a system from the viewpoint of Panama.²⁸ In Panama, although there does exist a national currency standard (the balboa), the balboa has been exactly equal in value to one U. S. dollar, and all the notes in circulation consist of U. S. dollar notes. Johnson states that such a system has the advantage of absolutely guaranteed stability of the currency in terms of the U. S. dollar, as against the disadvantage of resource transfer to the United States; dollars in circulation in Panama constitute an interest-free loan to the U. S. Government. This is the problem of seigniorage that always emerges from the international use of a national currency, and a sense of fairness calls for some form of compensation. Johnson is not in favor of the establishment of a central bank, reasoning that it would be more prone to issue more government debt than is prudent to maintain the stability of the currency. Instead, he proposes the establishment of a currency board, which would issue balboas in exchange for dollars and would invest the dollars so obtained in U. S. securities. According to Johnson, this is the better way of recouping most of the resource loss implied by the present arrangement, while maintaining stability and confidence in the currency.

When a number of economies approximately equal in size are involved in monetary unification, the use of a particular national currency as international money is not practicable from the viewpoint of either seigniorage or national prestige; an entirely new currency would have to be created.

Such is the case with European monetary unification. Triffin (1969), Cooper (1973a and 1973b), Ingram (1973), Johnson (1971 and 1973a), Mundell (1973b), Magnifico and Williamson (1972), Oppenheimer (1973), Griffiths (1971), Allen (1974), Pearce (1974), and Lamfalussy (1974) may be cited as representative of current thinking

²⁸ Mundell (1972) discusses the same problem, but from the viewpoint of the mother country. When there is a "mother country" and a "colony," the former can gain seigniorage by monetary expansion (inflation). Inflation causes a transfer of resources because it increases the spending of the residents of the mother country, including the increased imports from the colony.

on the monetary implications of developments in Europe. Earlier, the discussion focused on whether the Europeans might succeed in creating a new European unit of account, say the *Europa*, the value of which might be defined in terms of some standard unit. It was argued that the creation of a new unit might itself have only a symbolic meaning but would be an important step toward the practice of transferring funds in *Europas* by the authorities and eventually by commercial banks in the European Community as well. The notes denominated in this unit would also be issued and held by the public. This would require the establishment of a Bank of Europe to issue and control the *Europas*.

As to the question of the distribution of benefits of money creation derivable from the establishment of a supranational bank, Mundell proposes formation of a reserve pool with quotas and a European bank—much like a small version of the International Monetary Fund. Any benefit could then be distributed according to quotas.

The contributors to the literature have not elaborated much on the process of creating a new, single international currency and on the distribution of the resultant seigniorage. This reflects the fact that monetary unification, if it is to be carried out with less than perfect consolidation of national currencies and central banks, becomes technically complex and lacks the basic simplicity that is enjoyed by a nation under a single political jurisdiction.

IMPLICATIONS OF EUROPEAN MONETARY UNIFICATION

Apart from its implications for individual countries, monetary unification in Europe bears importantly on present and future world monetary arrangements.²⁹ It is widely known that one of the motivations for monetary unification was to create a strong European currency that could rival the dollar as an international currency, thereby countering inflationary pressures from the United States. It has been argued, particularly by Cohen (1963) and Mundell (1973b), that the Europeans can restore the independence of collective monetary control by surrendering individual national monetary sovereignty.

The special status of the dollar as the intervention and reserve currency imparts a certain asymmetry to the present world monetary arrangement. As Mundell (1973b) puts it, "low-powered money in the United States becomes high-powered money in Europe." This has sometimes undermined the capacity of the European countries to control

²⁹ Extensive discussions can be found in Cooper (1973a and 1973b). Implications of narrowing the exchange rate band were explored by Johnson (1973c).

their money supply, although the situation may have changed since many European countries have resorted to floating their currencies.

The move to monetary unification can be best understood as a counterweight to the growing influence of the U. S. dollar in the international monetary economy rather than as a response to internal dynamism in Europe. It may be welcome if it indeed rectifies the asymmetry and bias of the dollar-based system.

The internal desire for economic integration and the ultimate goal of political unification in Europe and the externally motivated desire of creating a monetary bloc rivaling the United States, however, are two entirely different motivations toward establishing a common European currency. On a number of occasions Johnson tried to draw attention to the inconsistency between these two motivations and implicitly predicted the breakdown of the attempt—specifically (1971, p. 379):

The strategy of inducing acceptance of the symbolic goal of a common currency as a means of forcing policy harmonization . . . assumes that the necessary preconditions for unified political decisions have not yet been established, but need to be secured by a species of political chicanery. The desire to confront the dollar with a rival currency of comparable power assumes that political agreement on both this objective and the means of achieving it already exist, and need only to be suitably organized through supranational institutions.³⁰

One could say that the danger involved in these conflicting assumptions was subsequently borne out.

Although these strictures on the proponents of monetary integration have been dominant among economists, Williamson (1974) remains an exceptional voice in advocating not to wait until economic and political conditions are ready, but to launch a new monetary initiative to complement structural unification in Europe.

CONCLUDING REMARKS

The theory of optimum currency areas provides important academic insights into the conditions under which payments adjustment through flexible exchange rates does not work well. It has also led to the finding of, or at least the recognition of the presence of, more precisely defined benefits and drawbacks of a common currency, and new aspects of policy coordination among countries on pegged exchange rates. The rapid development of joint floating of some European currencies and the Euro-dollar market—and, more recently, managed floating of major

³⁰ For similar views, see Hirsch (1972a), Claassen (1972), and Oppenheimer (1973).

currencies—have been the principal grounds for the revived and continued interest in the creation of an optimum currency area.

It has been shown, however, that several theories, identified here as the traditional approach, are useful but do not cover the many facets of the problem comprehensively enough. As an alternative, the cost-benefit approach, which takes the benefits and costs of a common currency explicitly into account, has been suggested.

Theoretical contributions aside, one would conclude that implications from the discussion of optimum currency areas for the real problem of the optimal world monetary arrangement have been meager. Strictly speaking, the theory of optimum currency areas applies to a small region and is not applicable for global monetary arrangements. For the global problem, global welfare considerations would have to be invoked, and such considerations are outside the framework of the optimum currency area approach. (Perhaps the term "optimum currency area" is a misnomer. To draw implications from the theory based on the calculation of regional or national interests for splitting the world into currency blocs would be to tax the theory too heavily.)

Some of the proponents of various criteria in the theory of optimum currency areas have tended to exaggerate the weight of key currencies, particularly the U. S. dollar. To Mundell and McKinnon, currencies of the small countries are too feeble and too inefficient to perform their duties as money and are in the process of being eroded by a few powerful currencies of large nations. McKinnon, moreover, is pessimistic about the capacity of the authorities to conduct stabilization policy in the small countries, as can be seen in his emphasis on trouble originating at home rather than abroad. A similar denigrating view on the monetary authority in the small countries is also held by Johnson.

If these views were correct, the discussion of optimum currency areas would indeed assume great practical importance. The author, however, is becoming increasingly skeptical of these views. It seems that, at least in the current world environment, the small countries are growing increasingly diversified in their trading relationships and, as a consequence, are less willing to peg to a single monetary bloc. Contrary to the views of Mundell and McKinnon, the world seems to be heading toward a situation in which small countries are more and more diversified and independent in their trading and financial relationships with large countries.

Following the move of major countries to general floating in the spring of 1973, there was a brief period in which a majority of small countries (particularly developing ones) favored a peg against a single currency. What these small countries are now increasingly realizing is

that when the exchange rates among major currencies are fluctuating among themselves—a fact of life which small countries have to accept—to peg to a single major currency is likely to jeopardize their trade with countries other than the country of the pegged currency. Therefore, the recent move to peg to a basket of currencies by a number of small countries can be viewed as a step toward enlightened and more rational economic management. Small countries are coming to the realization of the importance of having their own exchange rate policy with the concept of an effective exchange rate. From this follows the conclusion that the theory of optimum currency areas is primarily a scholastic discussion which contributes little to practical problems of exchange rate policy and monetary reform.

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